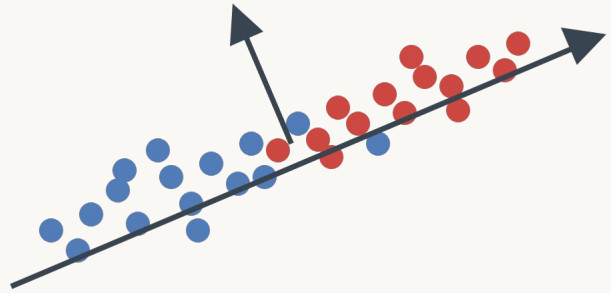
Look at the samples before testing them



Module 1 · Quality control in omics with PCA and sample–sample correlation

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AORN Santobono-Pausilipon IRCCS · BBCC2026 tutorial · TARGET-WT, 124 Wilms tumours

Every result inherits the quality of the samples

An omics matrix measures biology plus everything that happened to the sample, and a statistical test cannot tell the two apart.

- A technical effect touches thousands of genes at once.
- No single gene looks wrong. Only genome-wide summaries show the problem.
- Skip this step and an artefact can end up being called a biomarker.

Before Module 2, one question. Do the samples look the way we expect?

Each step of the experiment can leave a fingerprint

Collection → Extraction → Library → Sequencing → Quantification

Extraction: degradation, contamination

Library: kit, operator, batch

Sequencing: depth, lane, run date

Quantification: mapping rate, annotation

Level	Typical checks
Reads	base quality, adapters, duplication
Alignment	mapping rate, gene-body coverage
Samples	library size, PCA, correlation

When the batch follows the groups

$$\text{expression} = \mu + \beta \cdot \text{group} + \gamma \cdot \text{batch} + \varepsilon$$

μ	the gene's normal level	β	the biological effect
γ	a technical effect	ε	random noise

	β	γ	we observe
gene A	2	0	a difference of 2
gene B	0	2	a difference of 2

Principal component analysis

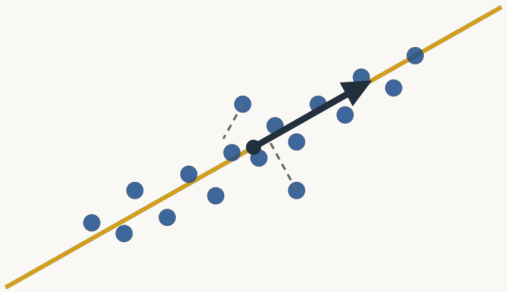
A change of coordinates (Pearson 1901, Hotelling 1933). The data are rotated onto new, uncorrelated axes ordered by the variance they carry, so a few axes can summarise thousands of genes.

In	Expression data
Out	scores for the samples, loadings for the genes, a variance for each axis
Not	a test, nor a clustering, and it never sees the group labels

In QC it answers one question. What are the largest sources of variation, and are they biological or technical?

A principal component is a weighted sum of genes

$$PC1 = w_1 \cdot gene_1 + w_2 \cdot gene_2 + \ldots + w_n \cdot gene_n$$



Each dot is a sample. The yellow line is PC1, the direction in which the samples spread the most.

Each sample gets a score, its expression of every gene multiplied by a weight and added up. That score is where the sample sits on the PCA plot.

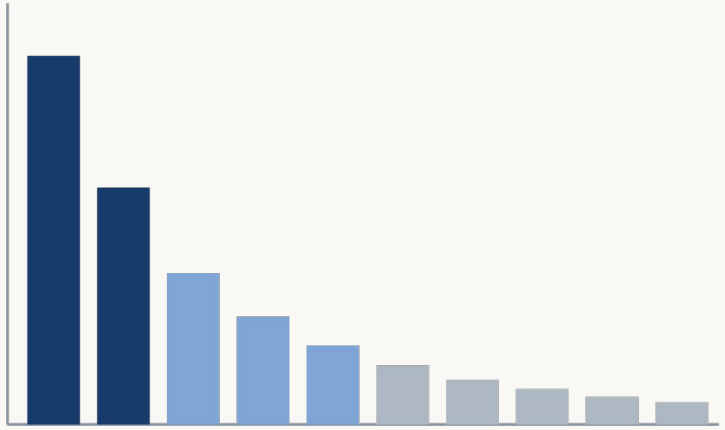
PCA chooses the weights so that the samples spread out as much as possible. PC2 does the same with what is left.

The weights are the loadings.

A gene with a large weight is a gene that drives that axis.

A final grade = 0.5 × written exam + 0.3 × oral exam + 0.2 × homework

How much of the picture each axis shows



A schematic scree plot, not this cohort. The real values are in the variance_explained table.

Each component carries a share of the total variation between samples. PC1 carries the most, then PC2, and so on.

The percentages on the axis labels, PC1 (x%) and PC2 (y%), say how much of the variation the plot actually shows.

```
summary(pca_result)$importance
```

Reading a PCA plot in QC

What you see	What it probably means	How to check
histology splits on a top PC	histology drives expression	are the loadings plausible?
clusters by site, run or date	a batch effect	colour by every metadata field
one sample far from all	degraded, contaminated or swapped	correlation, TPM sum
a smooth gradient	purity, stroma, degradation	top-loading genes
no split by histology	a smaller signal, not an absent one	PC3–PC5, Module 2

The loadings explain the scores. Mitochondrial, ribosomal or haemoglobin genes at the top of a PC usually mean a technical axis.

```
head(sort(abs(pca_result$rotation[, 1]),
          decreasing = TRUE), 20)
```

Sample–sample correlation

A number between −1 and +1 that says how well two profiles rise and fall together.

Computed for every pair of samples, it becomes a map of who resembles whom.

In	Normalized Expression matrix
Out	a n × n matrix, a clustered heatmap, a median r for each sample
Not	gene–gene co-expression, which correlates rows, and not a test

Samples of the same kind should resemble each other. One that resembles nobody is suspect.

How two samples move together

$$r = \mathrm{cov}(a, b) \,/\, (\mathrm{sd}(a) \cdot \mathrm{sd}(b))$$

The top, cov(a, b), asks whether the genes that are high in sample a are also high in sample b, and the low ones low.

Dividing by the two spreads puts the answer on a fixed scale, whatever the sequencing depth or the units.

−1	0	+1
opposite profiles	no relation	the same profile

In QC we compute it for every pair of samples, across all the expressed genes.

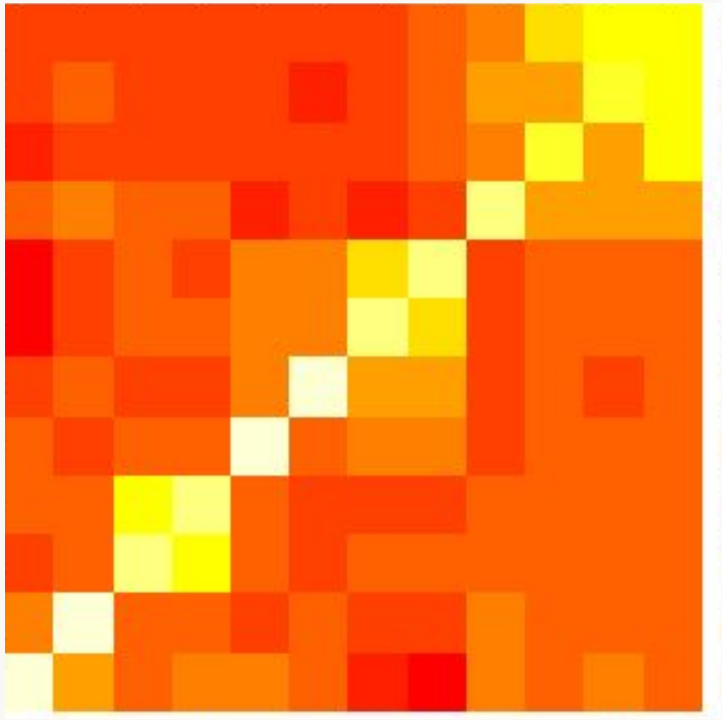
Why every r looks high

Housekeeping genes are high and many genes are low in every sample. That shared baseline alone pushes r between any two RNA-seq profiles towards 0.8–0.95, so judge r against the other samples, never against a fixed cut-off.

Pearson	A few extreme genes can move it.
Spearman	The same idea on ranks instead of values, so extreme genes matter less.

```
cor(expr_matrix, method = "pearson")
```

Reading the correlation heatmap



1. Samples that correlate well are close; the distance is 1 − r.
2. The closest samples are joined step by step into a tree, and rows and columns follow its order.
3. Light colors are groups. A dark stripe is a sample unlike the others, and a colour out of place in the top bar may be a mislabel.

Two views of the same data

	PCA	Correlation
looks at	the main axes of variation	how similar each pair of samples is
good at	batches, gradients, groups	single odd samples, mislabels
misses	effects beyond the first few PCs	differences hidden by the shared baseline
in R	prcomp()	cor(), pheatmap()

Both start from the same matrix, so they are two views rather than two independent tests. A sample flagged by both deserves a close look.

In paediatric data, every sample counts

Small cohorts make every sample count, pooling many sites makes batches likely, and every exclusion changes a small denominator. So it has to be documented.